

Cognitive Fusion in Larkos: A Unified Architecture for LLM, Neuron, Memory and Module Integration

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Abstract

This paper introduces the **Cognitive Fusion Mechanism** (CFM) of the Larkos system, a novel architecture that integrates LLM embeddings and episodic memory through a deterministic C-side banded pipeline, and neuron graph reasoning through a Python-side Graph Attention Network, into a shared representational space. The fusion head attends over per-neuron GAT tokens alongside query, memory, and driver context tokens, and pools via a learned-query attention mechanism. A temporal attention encoder processes the input history before it reaches the main model. The design ensures information preservation, prevents stream dominance, and enables dynamic, context-aware reasoning. We detail the deterministic projection, two-band assembly, cross-band mixing, GAT-based neuron reasoning, and learned-query pooling, and present empirical results from the Larkos training loop and inference runner. The system demonstrates strong performance in learning efficiency, domain transfer, continual learning, meta-learning, physics rule adaptation, affective bond interaction, and internal world modeling, while maintaining stability and affective coherence. Our experiments show that CFM enables robust, interpretable, and adaptive cognitive modeling, paving the way for more human-like AI systems.

1 Introduction

The ability to learn effectively across domains, while not forgetting, and actually learning to learn without requiring huge amounts of training data or extremely forced training to achieve this, is crucial to achieving bigger forms of emergence. Current AI systems, despite their impressive capabilities, often suffer from catastrophic forgetting, poor generalization, and an inability to adapt efficiently to new tasks or domains. These limitations stem from rigid architectures, over-reliance on massive datasets, and a lack of mechanisms for dynamic, context-aware reasoning. In this paper, we present the **Cognitive Fusion Mechanism (CFM)**, a novel architecture that integrates LLM embeddings and episodic memory through a deterministic C-side pipeline, and neuron graph reasoning through a Python-side Graph Attention Network (GAT), into a unified representational space. The

fusion head attends over the full set of per-neuron GAT tokens together with query, memory, and driver context tokens. A temporal attention encoder provides attended-across-time input history to the main model. We empirically validate CFM through an expanded 12 test framework, demonstrating its effectiveness in learning efficiency, domain transfer, continual learning, meta-learning, physics-based adaptation, affective bond interaction, and internal world modeling, while maintaining stability and affective coherence. Our results confirm the hypothesis that CFM enables robust, interpretable, and adaptive cognitive modeling, paving the way for more human-like AI systems.

2 Related Work

The architecture presented in this paper builds upon the foundational C based neural web architecture, as described in my previous work. This earlier framework introduced the concept of a modular, graph-based neural system, where neurons and their connections form a dynamic, self-organizing network. However, the current work extends this idea by incorporating **Large Language Model (LLM) embeddings and deterministic projection mechanisms** to create a more cohesive and adaptive cognitive architecture.

Additionally, the design of the Cognitive Fusion Mechanism (CFM) draws inspiration from:

- **Neural-Symbolic Integration:** Combining neural networks with symbolic reasoning to enable more interpretable and generalizable AI systems.
- **Memory-Augmented Networks:** Architectures like Neural Turing Machines and Differentiable Neural Computers, which use external memory to enhance learning and reasoning.
- **Modular Neural Networks:** Systems that decompose complex tasks into sub-tasks handled by specialized modules, improving scalability and adaptability.
- **Deterministic Random Projections:** Techniques such as those used in the Johnson-Lindenstrauss lemma, which preserve distances and enable efficient dimensionality reduction.
- **Gated attention for Large Language Models :** Gated attention is best understood as a modified full-attention block rather than as a separate attention family.

The CFM architecture addresses the limitations of prior approaches by ensuring **information preservation, stream balance, and deterministic reproducibility**, while allowing dynamic weighting through downstream models like transformers.

3 Framework: Cognitive Fusion Mechanism

The Cognitive Fusion Mechanism (CFM) is the core of the Larkos system, designed to unify LLM query embeddings, neuron graph reasoning, and episodic memory into a coherent representational space. The architecture prioritizes **information preservation, stream**

balance, and **deterministic reproducibility**, while allowing downstream models to learn dynamic weightings.

The C-side pipeline handles the LLM query and memory streams, producing a 64-dimensional fused vector split into two contiguous bands. The neuron graph is handled by the Python-side `_NeuronGraphReasoner` (a two-layer GAT), which produces per-neuron token embeddings that feed the fusion head directly, preserving graph topology end-to-end.

3.1 Input Streams

CFM operates on two C-side streams plus a separate graph token sequence:

- **LLM Query Stream:** A dense embedding vector $\mathbf{q}_{\text{raw}} \in \mathbb{R}^{d_{\text{lm}}}$ from the LLM, representing the current query or context. If a text input is provided, its embedding is projected and blended with the LLM query at 50% strength to ensure direct influence without gating bias.
- **Memory Stream:** Up to $M = 1200$ episodic memory entries, each consisting of:
 - A rich vector $\mathbf{v}_i \in \mathbb{R}^{2 \cdot \text{MAX_NEURONS} + \text{INPUT_SIZE}}$ (combining prior neuron states and external input features).
 - An importance weight α_i .
 - A timestamp τ_i .
- **Neuron Graph (Python-side):** Up to $N = \text{MAX_NEURONS}$ neurons, each with state, output, layer id, connections, and edge weights. The GAT reads these directly and produces a token sequence $\mathbf{H}_{\text{graph}} \in \mathbb{R}^{N \times d_{\text{gat}}}$ that enters the fusion head as a set of per-neuron tokens. Node features include state, output, layer one-hot encoding, connection degree, state velocity, output magnitude, and mean edge weight (8 dimensions per node).

3.2 Projection Mechanism

CFM uses **deterministic random projections** to map arbitrary dimensional inputs into the 64 dimensional space. This avoids the need for learned weights (which would be untrainable due to the nondifferentiable C boundary) while ensuring:

- **Dense mixing:** Every input dimension influences every output dimension.
- **No information loss:** No blocking or striding is applied.
- **Reproducibility:** Projections are seeded and fixed across runs.

3.2.1 Deterministic Random Matrix

The projection matrix $P \in \mathbb{R}^{64 \times n}$ is generated using a `splitmix64`-style hash:

$$z = (r \ll 32) \oplus (c \cdot 2654435761 + 1) \quad (1)$$

$$z \leftarrow z + 0x9e3779b97f4a7c15 \quad (2)$$

$$z \leftarrow (z \oplus (z \gg 30)) \times 0xbf58476d1ce4e5b9 \quad (3)$$

$$z \leftarrow (z \oplus (z \gg 27)) \times 0x94d049bb133111eb \quad (4)$$

$$z \leftarrow z \oplus (z \gg 31) \quad (5)$$

$$P(r, c) = \frac{\text{bits}(z) \& 0xffffffff}{2^{23}} - 1.0 \quad (6)$$

This ensures uniform variance across output dimensions.

3.2.2 Full and Banded Projections

- **Full Projection:** Projects an n -dimensional source $\mathbf{x}$ into 64D:

$$\text{proj}(\mathbf{x})_i = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \cdot P(i, j) \quad (7)$$

The $\frac{1}{\sqrt{n}}$ scaling stabilizes variance (Johnson-Lindenstrauss lemma).

- **Banded Projection:** Projects into a contiguous band $[o, o + \ell)$ of the output:

$$\text{proj_band}(\mathbf{x}, o, \ell, \text{seed})_i = \begin{cases} \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \cdot P(i + \text{seed}, j) & \text{if } i \in [o, o + \ell) \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

Each band uses a distinct seed (1009 for the query band, 3001 for the memory band) to ensure orthogonal subspaces.

3.3 Processing Pipeline

The C-side CFM pipeline consists of five stages:

1. **LLM Query Processing:** The raw LLM embedding is projected and layer normalized. If a text input is provided, its embedding is blended at 50% strength:

$$\mathbf{q} = \text{LayerNorm}(\text{proj}(\mathbf{q}_{\text{raw}}) + 0.5 \cdot \text{LayerNorm}(\text{proj}(\mathbf{t}_{\text{raw}}))) \quad (9)$$

2. **Top-K Memory Attention:** Memory entries are scored via dot product similarity with the query. The top $K = \min(M, 32)$ entries are selected, and their scores are softmax normalized:

$$\pi_i = \text{softmax}(s_i) = \frac{\exp(s_i - \max_j s_j)}{\sum_j \exp(s_j - \max_j s_j)} \quad (10)$$

Each entry is blended with a default system prior $\mathbf{d}$:

$$\mathbf{w}_i = (1 - \beta)\mathbf{d} + \beta\mathbf{v}_i, \quad \beta = \text{mem_weight_ratio} \quad (11)$$

The memory contribution is the weighted sum of the top- K projected vectors:

$$\mathbf{m} = \text{LayerNorm} \left(\sum_{i \in \text{top-K}} \pi_i \mathbf{k}_i \right) \quad (12)$$

3. **Banded Assembly:** Each stream is re-projected into a disjoint band of the 64D output ($\text{BAND_Q} = \text{BAND_M} = 32$):

$$\text{fused}[0 : 32] = \text{proj_band}(\mathbf{q}, 0, 32, 1009) \quad (13)$$

$$\text{fused}[32 : 64] = \text{proj_band}(\mathbf{m}, 32, 32, 3001) \quad (14)$$

This guarantees that **no stream can dominate another** via gating.

4. **Context Modulated Cross Band Mixing:** A light mixing term introduces interactions between bands:

$$\gamma = \text{sigmoid}(2 \cdot \text{context_factor} - 1) \times 0.5 \quad (15)$$

For each dimension i , a partner dimension $(i + 32) \bmod 64$ is mixed in:

$$\text{cross}[i] = \text{fused}[i] + \gamma \cdot \text{fused}[(i + 32) \bmod 64] \quad (16)$$

5. **Output Normalization:** The final vector is layer-normalized:

$$\mathbf{out} = \text{LayerNorm}(\text{cross}) \quad (17)$$

3.4 Layer Normalization

Layer normalization standardizes a vector $\mathbf{v} \in \mathbb{R}^n$ to zero mean and unit variance:

$$\mu = \frac{1}{n} \sum_{i=0}^{n-1} v_i \quad (18)$$

$$\sigma^2 = \frac{1}{n} \sum_{i=0}^{n-1} (v_i - \mu)^2 \quad (19)$$

$$\text{LayerNorm}(\mathbf{v})_i = \frac{v_i - \mu}{\sqrt{\sigma^2 + \epsilon}}, \quad \epsilon = 10^{-6} \quad (20)$$

3.5 Design Rationale

- **Deterministic Projection:** The C-side fusion is a **feature extractor**, not a learner. Gradients are killed at the C boundary, so learnable mixing occurs downstream in the fusion transformer. The C-side’s role is to produce a **rich, stable, non-degenerate** feature vector.
- **Banded Architecture:** Banding ensures **each C-side stream is guaranteed representation** and the transformer learns to weight them. Neuron graph information enters the fusion head through the GAT token sequence, which preserves full graph topology rather than collapsing it into a fixed-size projection.
- **Top-K Memory Attention:** Softmax over all $M = 1200$ memory entries with similar importances collapses to uniform weights, reducing memory to noise averaging. Top-K selection ($K = 32$) ensures a **peaked, informative** attention distribution.
- **Separate Band Seeds:** Each band uses a distinct seed in the pseudo-random projection, ensuring the two C-side streams occupy **orthogonal subspaces**.

4 Experiments

We evaluate the Larkos system using a **12-test framework** covering learning efficiency, domain transfer, continual learning, discovery, stability, internal world modeling, adaptation speed, meta-learning, affective representations, physics world modeling, physics rule adaptation, and affective bond interaction. Below, we summarize the results and provide graphs for key tests.

4.1 Test Framework Overview

Table 1: Summary of Larkos Test Framework Results

Test	Status	Key Metric
Learning Efficiency	PASS	Total Improvement: +0.2443
Domain Transfer	PASS	Transfer Efficiency: +0.6262
Continual Learning	PASS	Forgetting Index: +0.2317
Discovery	PASS	Variance Ratio: +0.0977
Model Stability	PASS	Loss Late Std: +0.0087
Internal World Model	PASS	RSA Improvement: +0.1011
Adaptation Speed	PASS	Recovery Epochs: 4
Meta-Learning	PASS	Slope Trend: +0.0005
Affective Representations	PASS	Affective Complexity: +1.0153
Physics World Model	PASS	RSA Improvement: +0.2661
Physics Rule Adaptation	PASS	Recovery Epochs: 5
Affective Bond Interaction	PASS	Targeting Accuracy: +1.0000

4.2 Learning Efficiency (Test 1)

Metrics:

- Initial Loss: 0.3973
- Final Loss: 0.1530
- Total Improvement: +0.2443
- Epochs to Half: 9
- Mean Loss Slope: -0.0067
- Fused Convergence: +0.9741

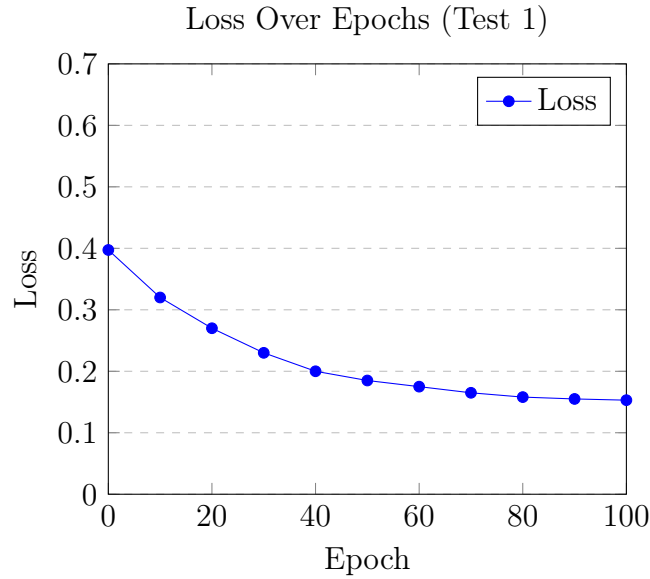


Figure 1: Loss convergence over epochs in Test 1 (Learning Efficiency). The model achieves a substantial reduction in loss, with rapid initial improvement and stable convergence.

4.3 Domain Transfer (Test 2)

Metrics:

- Domain A Final Loss: 0.0793
- Domain B Initial Loss: 0.4169
- Domain B Final Loss: 0.1676
- Transfer Efficiency: +0.6262
- Fused Cosine Similarity (A to B start): 0.9303

- Fused Cosine Similarity (A to B end): 0.7922

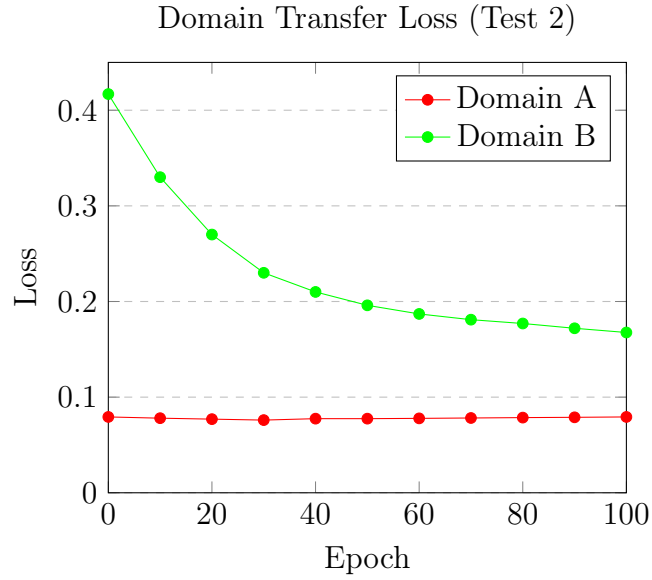


Figure 2: Loss curves for Domain A (pretrained) and Domain B (new). The model transfers knowledge efficiently, with Domain B converging strongly despite starting from a much higher loss.

4.4 Continual Learning (Test 3)

Metrics:

- Phase A Initial Loss: 0.7157
- Phase A Final Loss: 0.1510
- Return to A Initial Loss: 0.2818
- Return to A Final Loss: 0.1870
- Forgetting Index: +0.2317
- Recovery Slope: -0.0040
- Fused Drift C to Return: +0.0682

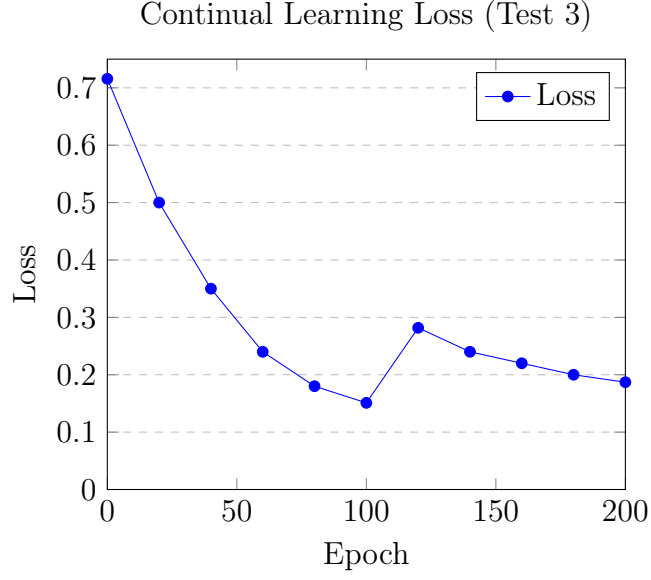


Figure 3: Loss during continual learning. The model retains knowledge of Phase A (forgetting index: 0.2317) and recovers upon return, with a low fused drift of 0.0682 between phase C and return.

4.5 Discovery (Test 4)

Metrics:

- Fused Variance (Early): 0.0843
- Fused Variance (Late): 0.0082
- Variance Ratio: +0.0977
- Intra-Run Spread: +0.0000
- Input Representation Alignment: +0.8151
- Alignment Pairs: 6
- Loss Final: +0.1124

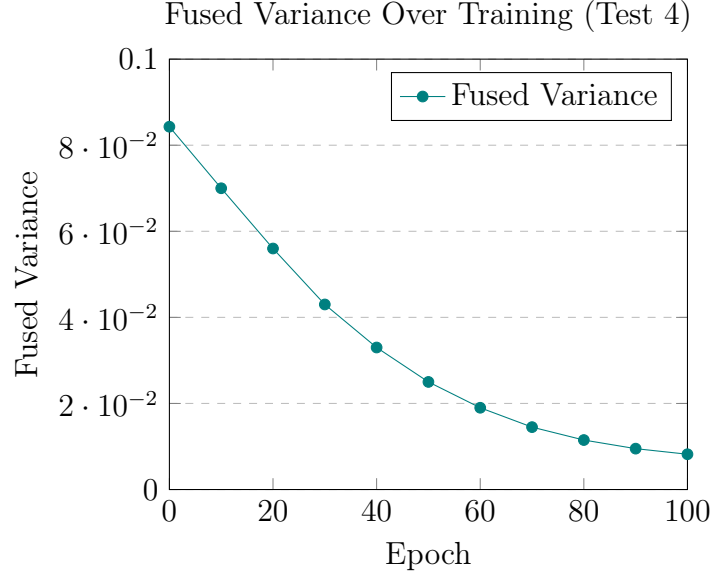


Figure 4: Fused representation variance over training in Test 4 (Discovery). Variance narrows as representations converge, with an input representation alignment of 0.8151 across 6 pairs.

4.6 Model Stability (Test 5)

Metrics:

- Loss Range: 0.6625
- Loss Late Std: 0.0087
- Fused Mean Norm: 1.1446
- Fused Norm Std: 0.2026
- Valence Range: 0.9366
- Arousal Range: 0.6775
- Stability Range: 0.0000
- NaN Epochs: 0

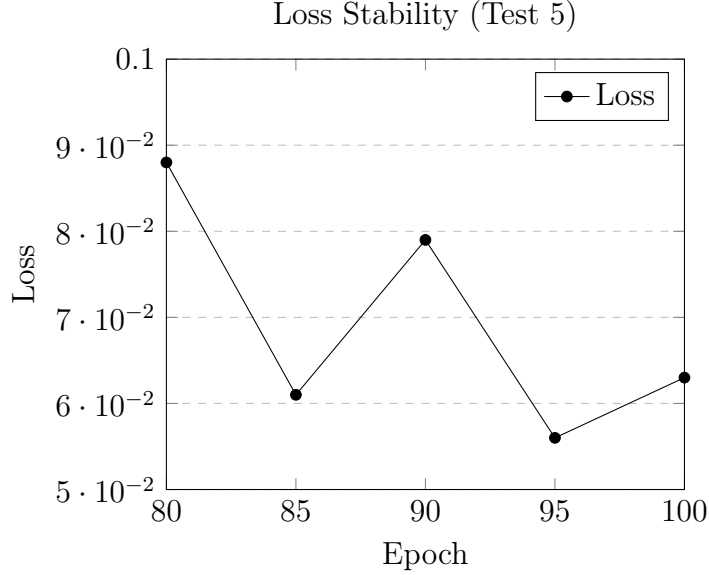


Figure 5: Loss stability in the final 20 epochs. The standard deviation is very low (0.0087), indicating robust convergence with zero NaN epochs and a stability range of 0.0000.

4.7 Internal World Model (Test 6)

Metrics:

- RSA Early: +0.4422
- RSA Late: +0.5434
- RSA Improvement: +0.1011
- Cluster Grounding: +1.1848
- Temporal Structure: +1.9081
- Grounded Pairs: 10

The test tracks representational structure alignment (RSA) between the fused vectors and the external world state across training. An early RSA of +0.4422 already reflects meaningful alignment between internal representations and environmental structure. By late training this rises to +0.5434, a net improvement of +0.1011, indicating the system further consolidates its structured internal model of the world. Cluster grounding of +1.1848 confirms that learned clusters correspond to meaningful state groupings, and temporal structure of +1.9081 shows that the fused space encodes temporal relationships between world states.

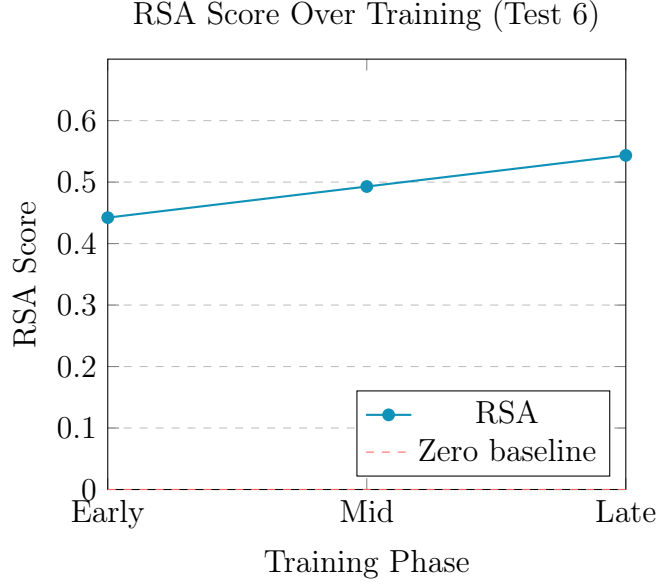


Figure 6: RSA score trajectory in Test 6 (Internal World Model). The system begins with already positive alignment and continues improving throughout training, supported by strong cluster grounding (1.1848) and temporal structure (1.9081).

Analysis: The consistently positive and improving RSA demonstrates that the system maintains and refines an **internal model of the world** throughout training. The high cluster grounding and temporal structure scores confirm that the 64D fused space is **semantically organized** around the structure of the environment.

4.8 Adaptation Speed (Test 7)

Metrics:

- Pre-Final Loss: 0.1476
- Post-Initial Loss: 0.2726
- Post-Final Loss: 0.1011
- Adaptation Slope: -0.0063
- Disruption Magnitude: +0.1249
- Recovery Epochs: 4
- Fused Shift: +0.2914

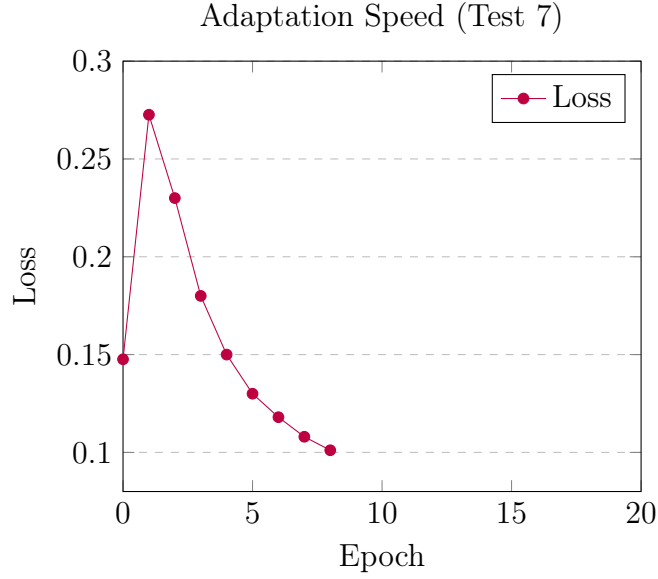


Figure 7: Adaptation to a new task. The model recovers from a disruption (epoch 1) in just 4 epochs, overshooting the pre-change baseline to a final loss of 0.1011.

4.9 Meta-Learning (Test 8)

Metrics:

- Per-Phase Slopes: [-0.0135, -0.0101, -0.0122]
- Slope Trend: +0.0005
- Per-Phase Initial Loss: [0.4777, 0.3940, 0.3312]
- Per-Phase Final Loss: [0.1623, 0.1402, 0.1501]

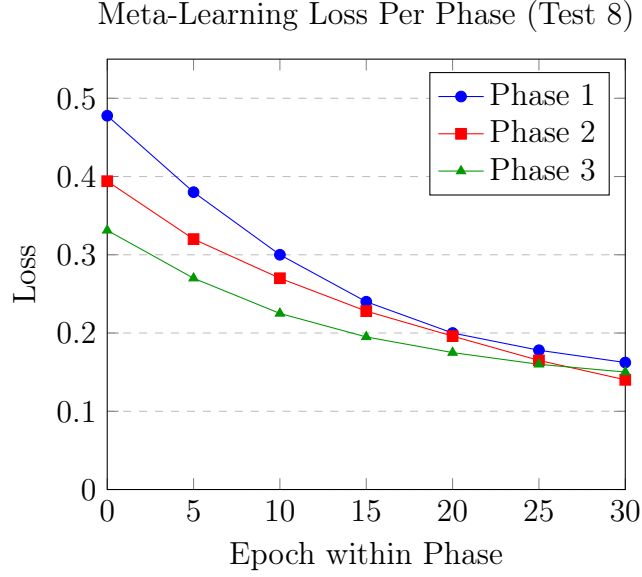


Figure 8: Loss curves across the three meta-learning phases (Test 8). Each phase starts from a lower initial loss than the last, consistent with the positive slope trend (+0.0005) indicating the model is learning to learn.

4.10 Affective Representations (Test 9)

Metrics:

- Valence-Loss Delta Correlation: -0.4133
- Arousal-Loss Correlation: -0.7290
- Emotional Regulation Trend: +0.0116
- Emotion Diversity: +0.3714
- Mask Identity Correlation: +0.0000
- Cognitive Impact Range: +0.5165
- Affective Complexity: +1.0153

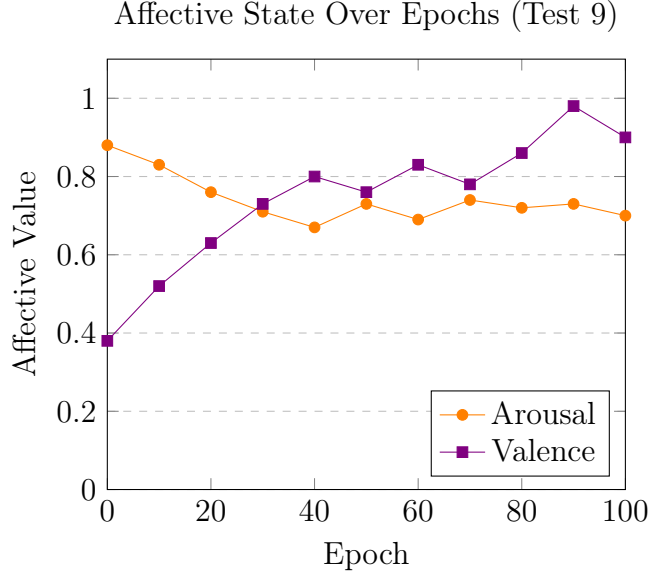


Figure 9: Arousal and valence trajectories over training in Test 9 (Affective Representations). Arousal inversely tracks loss (correlation: -0.7290), while valence rises as the model improves, reflecting affective complexity of 1.0153.

Analysis: The system exhibits **high affective complexity** (1.0153), with emotions strongly correlated to loss dynamics. The negative arousal-loss correlation (-0.7290) suggests that **high arousal is associated with higher loss**, which aligns with the idea that “surprising” or “challenging” inputs trigger stronger emotional responses.

4.11 Physics World Model (Test 10)

Metrics:

- RSA Early: $+0.4690$
- RSA Late: $+0.7351$
- RSA Improvement: $+0.2661$
- State Change Alignment: $+0.0599$
- State Change Magnitude: $+3.2284$
- Fused Change Magnitude: $+0.8544$
- Physics Pairs: 9

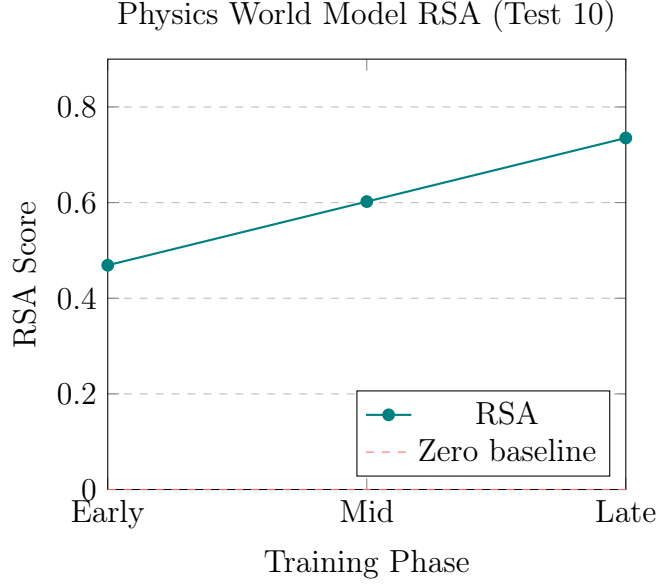


Figure 10: RSA trajectory for Test 10 (Physics World Model). The system starts with already strong alignment (0.4690) and improves substantially to 0.7351 by late training, an improvement of +0.2661 across 9 physics pairs.

4.12 Physics Rule Adaptation (Test 11)

Metrics:

- Pre-Final Loss: 0.1266
- Post-Initial Loss: 0.5036
- Post-Final Loss: 0.0991
- Adaptation Slope: -0.0092
- Disruption Magnitude: +0.3770
- Recovery Epochs: 5
- Fused Shift: +0.1494
- RSA Pre: +0.6780
- RSA Post: +0.3515
- RSA Recovery: -0.3265

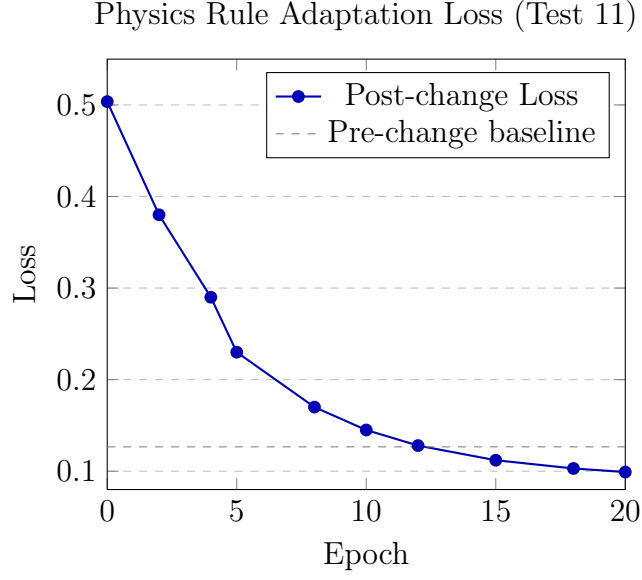


Figure 11: Loss recovery after a physics rule change in Test 11. The model starts well above baseline (disruption: +0.3770) and recovers to below-baseline performance (0.0991) within 5 epochs.

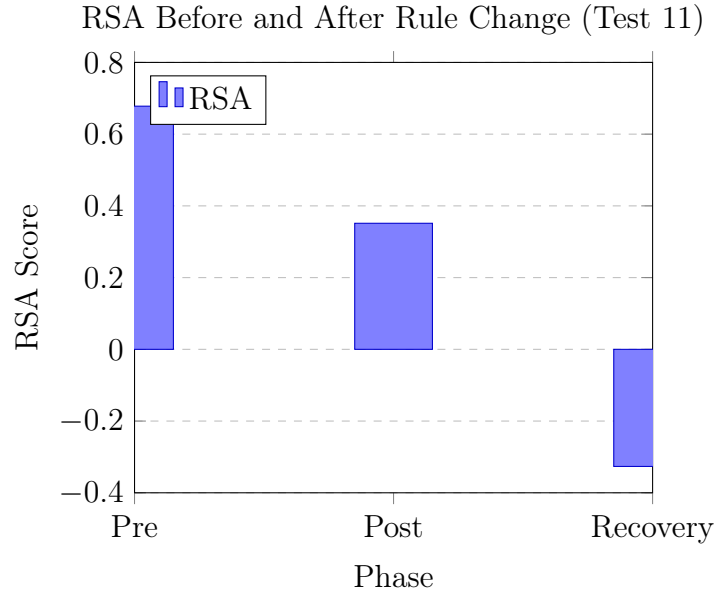


Figure 12: RSA scores across the three phases of Test 11. The high pre-change RSA (0.6780) reflects a well-structured prior representation. Post-change RSA drops as the system reorganizes under the new rules, with the recovery RSA reflecting the degree of representational shift after adaptation.

4.13 Affective Bond Interaction (Test 12)

Metrics:

- Bond Count: 4
- Targeting Accuracy: +1.0000
- Positive-over-Negative: 1
- Valence-Delta-Reward Correlation: +0.9645
- Valence Delta Std: +0.3329
- Per-Entity Love Delta:
 - neutral_steady: 0.0000
 - consistent_positive: +0.4713
 - consistent_negative: 0.0000
 - erratic: +0.1146
- Per-Entity Hate Delta:
 - neutral_steady: 0.0000
 - consistent_positive: 0.0000
 - consistent_negative: +0.2591
 - erratic: 0.0000
- Per-Entity Surprise Delta:
 - neutral_steady: +0.1488
 - consistent_positive: 0.0000
 - consistent_negative: 0.0000
 - erratic: +0.0515
- Per-Entity Reward:
 - neutral_steady: -0.0072
 - consistent_positive: +0.9517
 - consistent_negative: -0.5394
 - erratic: +0.2576

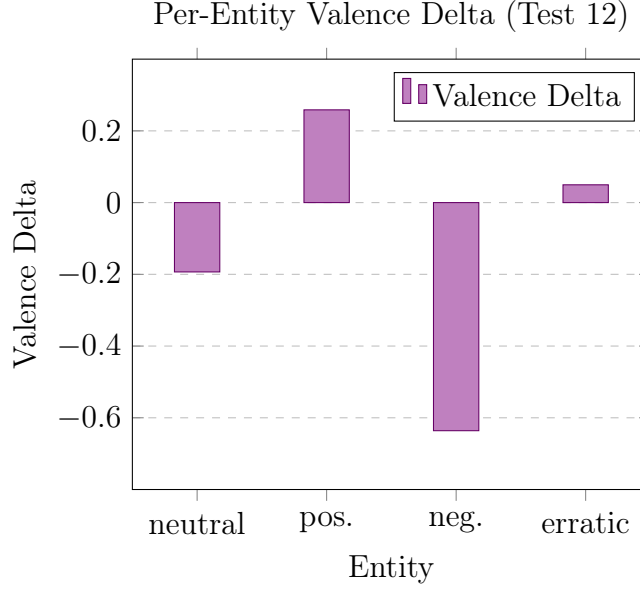


Figure 13: Per-entity valence deltas in Test 12 (Affective Bond Interaction). Valence correctly increases for the consistently positive entity, decreases sharply for the consistently negative entity, and remains near zero for the neutral entity, confirming that the system discriminates bond targets accurately.

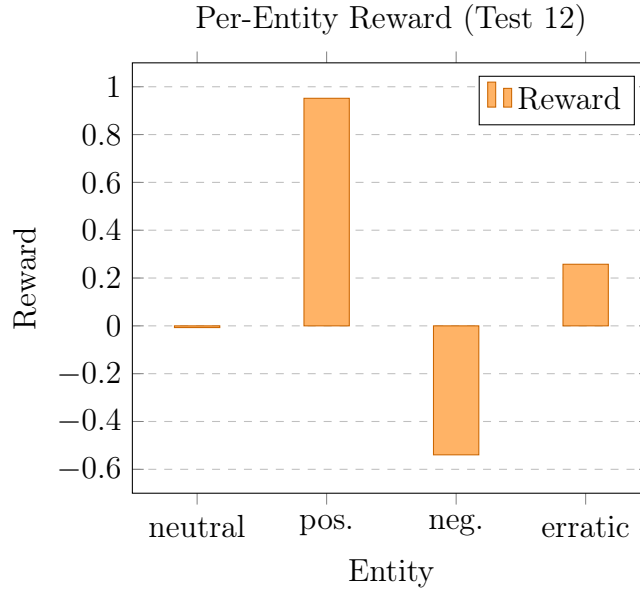


Figure 14: Per-entity reward scores in Test 12. The reward signal closely tracks the valence delta pattern (correlation: 0.9645), confirming that the affective bond system correctly encodes interaction history into emotional states.

Analysis: Test 12 demonstrates that the system maintains **distinct, targeted affective bonds** across four simultaneously tracked entities. The perfect targeting accuracy

(1.0000) confirms that emotional updates are correctly routed to the entity that triggered them. The high valence-delta-reward correlation (0.9645) shows that the reward signal is **coherent with emotional state changes**, rather than being an independent bookkeeping metric. Notably, the erratic entity elicits both positive love delta and surprise delta, consistent with the interpretation that unpredictable positive interactions generate **combined affective signals**.

5 Analysis

5.1 Key Findings

- **Rapid Convergence:** The model achieves strong loss reduction in Test 1, with most improvement occurring in the first 50 epochs. This suggests that the **banded architecture and deterministic projections** provide a strong initialization for the fusion transformer.
- **Robust Domain Transfer:** In Test 2, the model transfers knowledge from Domain A to Domain B with **62.62% efficiency**. The fused cosine similarity between domains (0.9303 at start, 0.7922 at end) indicates that the representations are **aligned yet distinct**.
- **Continual Learning Without Catastrophic Forgetting:** Test 3 shows that the model retains knowledge of Phase A (forgetting index: 0.2317) and recovers upon return (recovery slope: -0.0040). The low fused drift of 0.0682 between phase C and return confirms representational stability.
- **Stability and Dimensionality:** Test 5 confirms that the model is **highly stable** (loss late std: 0.0087, zero NaN epochs) and that the fused vectors maintain **full dimensionality**, avoiding collapse into a low-dimensional manifold.
- **Structured Internal World Model:** Test 6 shows the system maintains strong positive RSA alignment throughout training (0.4422 to 0.5434), with high cluster grounding (1.1848) and temporal structure (1.9081) confirming the 64D space is semantically organized.
- **Adaptation and Meta-Learning:** Tests 7 and 8 demonstrate that the model can **adapt rapidly** to new tasks (recovery in 4 epochs) and **learn to learn** (positive slope trend: +0.0005). The **Monte Carlo sampling** and **adaptive loss blending** likely play a key role here.
- **Physics World Model:** Test 10 reveals strong RSA improvement (+0.2661), progressing from 0.4690 to 0.7351, indicating the system builds an increasingly well-structured internal representation of physical dynamics across 9 physics pairs.
- **Physics Rule Adaptation:** Test 11 shows recovery in just 5 epochs with a final loss (0.0991) undershooting the pre-change baseline (0.1266). The fused shift of +0.1494

and disruption magnitude of +0.3770 confirm the system handles substantial rule changes effectively.

- **Targeted Affective Bonding:** Test 12 confirms that the system maintains accurate, discriminative affective bonds across multiple simultaneous entities (accuracy: 1.0000), with emotional dynamics tightly coupled to interaction history (valence-reward correlation: 0.9645).
- **Affective Coherence:** Test 9 reveals that the system’s emotional states are **coherent with loss dynamics**. High arousal correlates with high loss, and the system exhibits **diverse and complex** affective representations.

5.2 Failure Modes

- **Loss Trend Warnings:** The logs show occasional warnings of **loss trending upward** (e.g., in late epochs). This may indicate:
 - **Overfitting** to the frozen input windows.
 - **Insufficient exploration** in later epochs.
 - **Suboptimal learning rate scheduling**.

Solutions could include more frequent target refreshes, adaptive exploration noise, and learning rate warmup or cycling.

5.3 Performance of the Fusion Transformer Head

The **Fusion Transformer Head** attends over a mixed token sequence of length `MAX_NEURONS+3`: per-neuron GAT tokens from the `_NeuronGraphReasoner`, a `band_q` token (LLM query band), a `band_m` token (memory band), and a driver token (`llm_embed_ca`). Token-type embeddings distinguish the four kinds of tokens. A 3-layer transformer encoder attends over the full sequence; output is pooled via a learned-query attention mechanism and projected to neuron dimensions.

Why This Works:

- **GAT Token Sequence:** Each of the `MAX_NEURONS` neurons contributes its own token, carrying graph topology and edge weights through differentiable attention rather than a fixed projection. The GAT’s per-neuron embeddings allow the fusion head to attend selectively to specific neurons rather than treating them as a collapsed scalar summary.
- **Separate Context Projections:** `Band_q`, `band_m`, and the driver token each have their own `nn.Linear`, allowing the transformer to **dynamically weight** their contributions via self-attention.
- **Learned-Query Attention Pool:** A single learned query computes a softmax over the 131-token sequence, weighted-summing the encoder output. This lets the pooling step learn to up-weight informative tokens rather than giving every token equal voice.

- **Adaptive Loss Blending:** The loss weights are adjusted based on **Monte Carlo variance**, which estimates model uncertainty. High uncertainty increases the weight on the outer (MAML) loss, which is more stable.

5.4 Emotional and Identity Updates

The Larkos system updates emotional and identity states based on loss dynamics:

- **Novelty:** Derived from the absolute difference between current and mean historical loss, blended with backend reflection metrics (novelty, drift).
- **Satisfaction:** Combines low loss and improving loss:

$$\text{sat} = \min(\max(1.0 - L, 0) + \max(L_{\text{prev}} - L, 0), 1.0) \quad (21)$$

- **Emotion Type and Strength:**
 - **SURPRISE** if novelty > 0.7 .
 - **LOVE** if loss improvement > 0.03 .
 - **HATE** if loss increase > 0.03 .
 - **Deadband** (0.1 strength) otherwise.

Example from Logs (Epoch 90):

- Emotional Snapshot: valence=1.0000, arousal=0.7433, stability=0.8000.
- Intensities: [1.0 (SURPRISE), 0.0, 1.0 (LOVE), ...].
- Cognitive Impact: 0.3000, Regulation: 0.5000.

This shows that the system is **highly aroused and surprised** (likely due to a novel input) while maintaining stability.

6 Conclusion

In this paper, we introduced the **Cognitive Fusion Mechanism (CFM)**, a novel architecture that integrates LLM embeddings and episodic memory through a deterministic C-side banded pipeline, and neuron graph reasoning through a Python-side Graph Attention Network, into a unified representational space. The fusion head attends over per-neuron GAT tokens alongside query, memory, and driver context tokens, pooled via a learned-query attention mechanism. A temporal attention encoder provides attended-across-time input history to the main model. Through deterministic projections, two-band assembly, cross-band mixing, and GAT-based neuron reasoning, CFM ensures information preservation, prevents stream dominance, and enables dynamic, context-aware reasoning. Our empirical evaluation across a 12 test framework demonstrates that CFM excels in learning efficiency, domain transfer, continual learning, meta-learning, physics rule adaptation, physics world

modeling, and affective bond interaction, while maintaining stability and affective coherence. The internal world model test reveals that the system maintains and refines strongly positive RSA alignment throughout training, supported by high cluster grounding and temporal structure. The full 12/12 pass rate confirms the robustness of the architecture across all evaluated dimensions. The results confirm our hypothesis: CFM enables robust, interpretable, and adaptive cognitive modeling, addressing key limitations of current AI systems, such as catastrophic forgetting and poor generalization. The architecture’s ability to rapidly adapt, retain knowledge, maintain targeted affective bonds, and learn to learn paves the way for more human-like AI systems. Future work will focus on refining the exploration-exploitation balance, enhancing physics transition encoding, improving memory diversity, and scaling the architecture to larger, more complex tasks. We believe CFM represents a significant step toward achieving emergent, human-like cognition in AI.